SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

AgroLink Ethiopia

Inclusive Agricultural Marketplace and Advisory Platform

Document Version: 1.0

Document Status: Proposed — For Review and Approval

Project Type: Web and Mobile-Based Agricultural Information System

Target Country: Ethiopia

Web Application: React.js

Mobile Application: React Native

Backend: Node.js and Express.js

Database: PostgreSQL

Architecture: Client–Server Architecture with RESTful APIs

Development Methodology: Iterative and Incremental Development

Primary Languages: English and Amharic (with localization support)

AI/ML Integration: Not included in this release

Document Control

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Document	Software Requirements Specification
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Version	Description
1.0	Initial web-based agricultural marketplace and advisory specification
2.0	Revised for web and mobile applications, five main actors, assisted farmer access, and removal of transportation management

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements of AgroLink Ethiopia, an inclusive agricultural marketplace and advisory platform designed to connect farmers, buyers, agricultural service providers, and support personnel.

The system will provide two primary access channels:

- A web application accessible through desktop, tablet, and mobile browsers.
- A mobile application developed with React Native for Android and, where supported, iOS devices.

The platform will support two different farmer access methods:

1. Direct farmer access: digitally experienced farmers use the mobile application or website independently.
2. Assisted farmer access: traditional farmers receive assistance from authorized Support Agents who help them register, publish products, and use platform services.

The specification defines the system's actors, requirements, business rules, user workflows, database expectations, application architecture, security controls, testing requirements, and acceptance criteria.

It will serve as the primary reference for system analysis, design, development, testing, deployment, and project evaluation.

1.2 Problem Statement

Agricultural producers in Ethiopia may experience difficulties when trying to access buyers, publish available products, compare agricultural prices, find agricultural inputs, and obtain professional agricultural guidance.

Some farmers have access to smartphones and digital services, while others may have limited internet access, limited digital literacy, or no personal smartphone.

Traditional farmers may therefore face barriers when using digital agricultural marketplaces, even when the marketplace could help them reach more customers.

Other challenges include:

- Agricultural product information may be distributed across different communication channels.
- Farmers may have difficulty finding potential buyers.
- Buyers may lack convenient access to structured information about available agricultural products.
- Product prices, quantities, and availability may not be consistently presented.
- Farmers may have difficulty accessing agricultural inputs.
- Farmers may need professional agricultural consultations.
- Traditional farmers may require assistance creating digital product listings.
- Manual records can make it difficult to track orders and previous transactions.
- Communication between farmers, buyers, and agricultural service providers may be fragmented.

AgroLink Ethiopia proposes a centralized digital platform that connects agricultural participants while providing an assisted access model for farmers who cannot independently use digital devices.

1.3 Project Objectives

1.3.1 General Objective

To design and develop an inclusive web and mobile agricultural marketplace and advisory platform that connects farmers, buyers, and agricultural service providers in Ethiopia while supporting both direct digital access and assisted access for traditional farmers.

1.3.2 Specific Objectives

The system shall aim to:

1. Develop a centralized agricultural marketplace.
2. Allow modern farmers to register and publish agricultural products directly.
3. Allow authorized Support Agents to help traditional farmers create and manage product listings with consent.
4. Ensure that farmers remain the owners of their products and listings.
5. Enable buyers to search, filter, compare, and order agricultural products.
6. Enable agricultural input suppliers to publish seeds, fertilizers, tools, and other agricultural inputs.
7. Allow agricultural experts to provide human consultations.
8. Provide market information and agricultural educational resources.
9. Provide secure user registration, authentication, and role-based authorization.
10. Provide order management and transaction-history functionality.
11. Provide messaging and notifications.
12. Support user reviews, complaints, and dispute-management processes.
13. Provide administrative dashboards and operational reports.
14. Support English and Amharic interfaces.

15. Develop a responsive web application and a React Native mobile application.
16. Protect personal information and commercial records.
17. Improve access to agricultural marketplace services for farmers with different levels of digital literacy.

1.4 Intended Audience

This document is intended for:

- Project supervisors and evaluators.
- Software developers.
- UI/UX designers.
- Database designers.
- Software testers.
- System administrators.
- Farmers and agricultural stakeholders.
- Buyers.
- Agricultural input suppliers.
- Agricultural experts.
- Support Agents.
- Platform administrators.

1.5 Definitions and Requirement Conventions

Term	**Meaning**
Shall	Mandatory requirement
Should	Recommended requirement
May	Optional requirement
FR	Functional Requirement
NFR	Non-Functional Requirement
BR	Business Rule

AC	Acceptance Criterion
API	Application Programming Interface
RBAC	Role-Based Access Control
KYC	Know Your Customer / identity verification process
CRUD	Create, Read, Update, Delete
REST	Representational State Transfer
SRS	Software Requirements Specification

Each requirement shall have a unique identifier to support implementation, testing, and traceability.

1.6 Scope of This Specification

This specification covers:

- Web application.
- Mobile application.
- Backend REST API.
- PostgreSQL database.
- User registration and authentication.
- Farmer direct access.
- Assisted farmer onboarding.
- Agricultural product marketplace.
- Agricultural input marketplace.
- Buyer and order management.
- Agricultural consultations.
- Market information.
- Educational articles.
- Messaging and notifications.
- Support ticket management.
- Reviews and disputes.
- Administrative management.

- Security and authorization.
- Testing and deployment.

2. Project Overview and Scope

2.1 Product Description

AgroLink Ethiopia is a digital agricultural platform designed to connect agricultural producers with potential buyers and relevant service providers.

The platform will allow farmers to publish agricultural products, buyers to discover and order products, input suppliers to advertise agricultural inputs, and agricultural experts to provide consultations.

The system will also introduce an Assisted Farmer Service. Through this service, an authorized Support Agent can help farmers who do not own smartphones or lack the digital skills required to use the application independently.

All users will interact with a centralized backend and database.

The platform will provide consistent information across the web and mobile applications.

2.2 Target Users

The platform is designed for:

- Traditional farmers who require assistance.
- Modern farmers who can use smartphones or computers.
- Agricultural product buyers.
- Agricultural input suppliers.
- Agricultural experts.
- Support Agents.
- Platform administrators.

2.3 Features Included in Scope

The first release shall include:

1. User registration and login.
2. Role-based access control.
3. User profile management.
4. Farmer onboarding and profile management.
5. Assisted farmer registration.
6. Assisted product-listing creation.
7. Farmer-owned agricultural product listings.
8. Agricultural input listings.
9. Product search and filtering.
10. Product details and availability.
11. Shopping cart or order creation.
12. Order management and order history.
13. Agricultural consultation requests.
14. Agricultural expert profiles.
15. Market-price information.
16. Educational articles and resources.
17. In-platform messaging.
18. Notifications.
19. Support ticket management.
20. Reviews and ratings.
21. Complaints and disputes.
22. User and listing moderation.
23. Administrative dashboards.
24. Operational reports.
25. Audit logs.
26. English and Amharic localization support.
27. Responsive web application.
28. React Native mobile application.

2.4 Features Excluded from Scope

The following features are excluded from this release:

- ✓ Transportation-provider registration.
- ✓ Transporter dashboards.
- ✓ Transport booking.
- ✓ Delivery assignment.
- ✓ Delivery tracking.
- ✓ Vehicle management.
- ✓ Driver management.
- ✓ Delivery-fee calculation.
- ✓ Artificial intelligence and machine learning.
- ✓ AI chatbots.
- ✓ Automated crop-disease diagnosis.
- ✓ Automated market-price prediction.
- ✓ AI-generated agricultural recommendations.
- ✓ Automated credit scoring.
- ✓ Loan approval.
- ✓ Unapproved banking or mobile-money integrations.
- ✓ Guaranteed physical inspection of all products.
- ✓ Automatic government registration or regulatory certification.
- ✓ Automated negotiation between buyers and sellers.

Buyers and sellers may independently arrange physical transportation outside AgroLink. The platform will not manage or guarantee that arrangement.

2.5 Expected Outcomes

The system is expected to:

- ✓ Improve the discoverability of agricultural products.
- ✓ Connect traditional farmers with digital marketplace opportunities.

- ✓ Reduce barriers to digital participation.
- ✓ Improve communication between buyers and farmers.
- ✓ Make agricultural input information easier to access.
- ✓ Improve access to human agricultural consultations.
- ✓ Organize order and transaction records.
- ✓ Provide structured agricultural market information.
- ✓ Improve administrative monitoring and accountability.

These are expected benefits rather than guaranteed financial or agricultural outcomes.

2.6 Project Success Measures

The project should be evaluated through:

- ➢ Successful user registration and login.
- ➢ Successful farmer direct registration.
- ➢ Successful assisted farmer registration.
- ➢ Correct recording of farmer consent.
- ➢ Correct ownership of assisted product listings.
- ➢ Successful product publication and moderation.
- ➢ Successful product search and filtering.
- ➢ Successful order creation and processing.
- ➢ Successful consultation requests.
- ➢ Correct role-based authorization.
- ➢ Successful web and mobile API communication.
- ➢ Correct report generation.
- ➢ Successful security tests.
- ➢ Successful database backup and restoration.

3. Overall System Description

3.1 Product Perspective

AgroLink Ethiopia shall use a centralized client-server architecture.

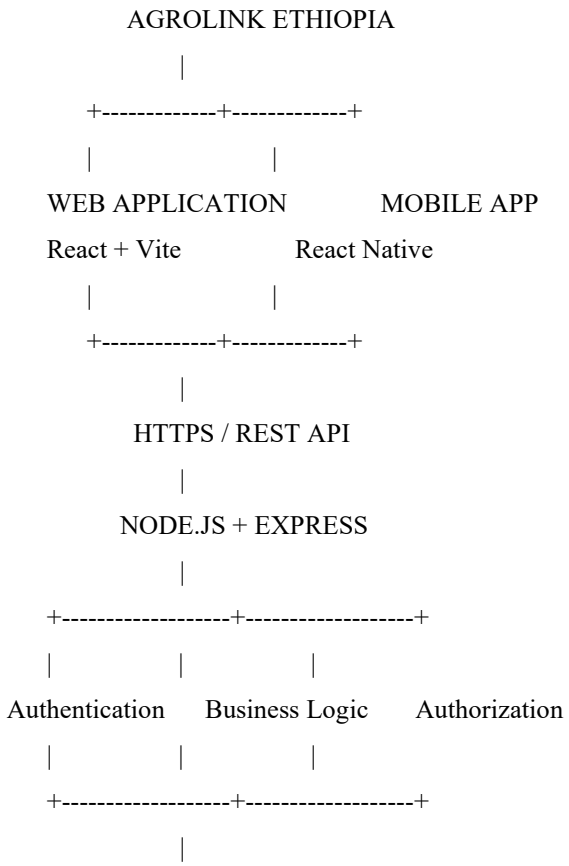
The system will consist of:

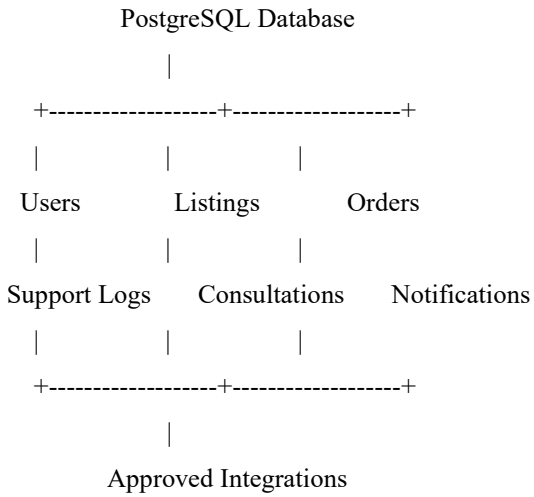
- Web client.
- Mobile client.
- Backend API.
- PostgreSQL database.
- Approved external services where required.

The web and mobile applications will communicate with the same backend API.

The backend will validate requests, authenticate users, enforce permissions, execute business rules, and access the database.

3.2 High-Level System Architecture





3.3 Operating Environment

Web Application

The web application shall support current stable versions of:

- Google Chrome.
- Microsoft Edge.
- Mozilla Firefox.
- Safari where available.

The interface shall support desktop, tablet, and mobile browser screen sizes.

Mobile Application

The mobile application shall be developed using React Native.

Android shall be the primary mobile target for the first release. iOS support may be included where the development and testing environment allows.

The mobile application shall communicate with the same backend API used by the web application.

Backend Environment

The backend shall use:

- Node.js.
- Express.js.
- JavaScript.
- PostgreSQL.
- RESTful APIs.
- Secure environment configuration.

Development Environment

Recommended development tools include:

- Visual Studio Code.
- Git and GitHub.
- Node.js and npm.
- PostgreSQL.
- Postman or an equivalent API testing tool.
- React Testing Library.
- Vitest or an approved test runner.
- Jest or an approved React Native testing approach.
- Supertest.
- Approved deployment services.

3.4 Assumptions

The system assumes that:

1. Some farmers have smartphones and internet access.
2. Some farmers will require assisted access through Support Agents.
3. Authorized Support Agents will be available to assist farmers.
4. Farmers will provide accurate product and personal information.
5. The project owner will define identity and account-verification procedures.

6. Agricultural experts will be responsible for the accuracy of their professional advice.
7. Buyers and sellers will agree on commercial terms through the approved order workflow.
8. Administrators will supervise Support Agent activities.
9. The project will provide approved hosting and database infrastructure.
10. Internet access will be available when users access online system functions.

3.5 Dependencies

The project depends on:

- React and Vite for the web application.
- React Native for the mobile application.
- Node.js and Express.js.
- PostgreSQL.
- Secure hosting.
- HTTPS.
- Image storage.
- Email or SMS services if selected.
- Qualified human agricultural experts.
- Trained Support Agents.
- Database backup and monitoring facilities.

3.6 System Limitations

AgroLink cannot guarantee the accuracy of information entered by farmers, buyers, suppliers, or third parties.

The platform will provide reporting and moderation mechanisms to address incorrect or suspicious information.

AgroLink will not own the products published by farmers and suppliers.

The system will not act as a transportation company or manage delivery services in this release.

Agricultural advice will be provided by human experts rather than an automated AI system.

4. Stakeholders and User Classes

4.1 Primary Actors

AgroLink Ethiopia shall use five principal application actors.

4.1.1 Farmer

The Farmer is an agricultural producer who uses the platform to publish agricultural products, manage orders, and access agricultural information and consultations.

Farmers may access the platform directly or through assisted support.

Direct Farmer Responsibilities

A digitally experienced farmer shall be able to:

- Register independently.
- Log in and log out.
- Manage a personal profile.
- Create agricultural product listings.
- Upload product images.
- Update product quantity and price.
- Activate or deactivate listings.
- View incoming orders.
- Accept or reject eligible orders.
- View order history.
- Request agricultural consultations.
- Read agricultural articles.
- View market-price information.
- Receive notifications.
- Communicate with buyers.

- Submit support requests.
- View relevant transaction records.

Assisted Farmer Responsibilities

A traditional farmer who cannot independently access the application shall be able to:

- Provide information to a Support Agent.
- Give consent for account registration.
- Provide agricultural product details.
- Confirm product quantity and price.
- Approve product publication.
- Receive information about incoming orders.
- Request corrections to product information.
- Withdraw permission for future assisted actions.

The farmer remains the legal and commercial owner of the farmer's account and agricultural product listings.

4.1.2 Buyer

A Buyer is an individual or organization interested in purchasing agricultural products.

The Buyer shall be able to:

- Register and log in.
- Manage a buyer profile.
- Search agricultural products.
- Filter products by category, location, price, and availability.
- View product details.
- View permitted seller information.
- Add products to a shopping cart where supported.
- Create orders.
- View order status.

- View order history.
- Communicate with relevant sellers.
- Cancel orders where permitted.
- Submit eligible reviews.
- Submit complaints and disputes.
- Receive notifications.
- Create support tickets.

4.1.3 Service Provider

The Service Provider is a generalized actor representing two types of agricultural service providers:

- Input Supplier.
- Agricultural Expert.

Each provider shall have a provider type that determines the permitted functions.

A. Input Supplier

An Input Supplier sells agricultural inputs such as:

- Seeds.
- Fertilizers.
- Agricultural tools.
- Equipment.
- Other approved agricultural supplies.

The Input Supplier shall be able to:

- Manage a supplier profile.
- Create input-product listings.
- Update product details.
- Set prices and quantities.

- Upload images.
- Manage available stock.
- View relevant orders.
- Respond to customer inquiries.
- Receive reviews and complaints.
- Receive notifications.

B. Agricultural Expert

An Agricultural Expert provides human agricultural consultation services.

The Agricultural Expert shall be able to:

- Create a professional profile.
- Publish areas of expertise.
- Provide professional background information.
- Define consultation availability.
- Receive consultation requests.
- Accept or reject requests.
- Reschedule consultations.
- Communicate with assigned farmers.
- Record consultation status.
- Maintain consultation history.
- Receive feedback.
- Receive notifications.

The two provider types shall not automatically share all permissions.

4.1.4 Support Agent

The Support Agent helps users access and use AgroLink, especially farmers who lack smartphones or digital skills.

The Support Agent shall be able to:

- Register farmers through an authorized workflow.
- Search for existing farmer accounts.
- Update permitted farmer profile information.
- Help farmers create agricultural product listings.
- Help farmers update existing listings.
- Record farmer consent.
- Record assistance history.
- Help farmers understand orders.
- Assist farmers in submitting consultation requests.
- Help farmers access market information.
- Create support tickets.
- Respond to assigned support tickets.
- Escalate disputes to the Administrator.

The Support Agent shall not:

- Access or request a farmer's password.
- Access a farmer's one-time verification code.
- Become the owner of a farmer's listing.
- Change a farmer's price or quantity without the required approval.
- Accept or reject commercial orders without explicit authorization.
- Withdraw money or approve payments.
- Access unrelated private user information.
- Change administrator permissions.
- Delete audit records.

All assisted actions shall be traceable.

4.1.5 Administrator

The Administrator is responsible for the overall operation and management of AgroLink.

The Administrator shall be able to:

- Manage user accounts.
- Assign and manage authorized roles.
- Review account verification.
- Manage Support Agent accounts.
- Monitor assisted farmer registrations.
- Review consent and assistance records.
- Moderate farmer listings.
- Moderate input listings.
- Manage product categories.
- Manage agricultural articles.
- Manage market-price information.
- Review agricultural expert profiles.
- Monitor consultations.
- Monitor orders.
- Review complaints and disputes.
- Manage support tickets.
- View administrative reports.
- Review audit logs.
- Configure platform settings.
- Manage approved notification settings.
- Suspend or reactivate accounts according to policy.

4.2 Actor Summary

Actor	**Main responsibility**
Farmer	Agricultural product owner and marketplace participant
Buyer	Agricultural product customer
Service Provider	Agricultural input supplier or agricultural expert
Support Agent	Assisted access and user support
Administrator	Platform management and governance

4.3 User Access Matrix

Function	Farmer	Buyer	Service Provider	Support Agent	Admin
Manage own profile	Yes	Yes	Yes	Yes	Yes
Create farmer listing	Own account	No	No	Assisted only	Authorized moderation
Manage own farmer listings	Yes	No	No	With consent and permission	Authorized moderation
Create input listings	No	No	Supplier only	No	Authorized moderation
Manage consultations	Request	Request if allowed	Expert only	Assist with consent	Oversight
Search products	Yes	Yes	Yes	Yes	Yes
Place orders	Where applicable	Yes	Where applicable	Assisted only	Authorized testing
Manage own orders	Yes	Yes	Yes	Assistance only	Oversight
Create support tickets	Yes	Yes	Yes	Yes	Yes
Manage support tickets	Own	Own	Own	Assigned	All authorized
Moderate listings	No	No	No	Report/escalate	Yes
Manage user roles	No	No	No	No	Yes
View global reports	No	No	No	Limited	Yes
View audit logs	No	No	No	Own permitted activity	Yes

Exact authorization shall be enforced by the backend.

5. System Features and Functional Requirements

5.1 User Registration and Authentication

Feature ID: FEAT-AUTH

- ✓ FR-AUTH-001: The system shall allow users to register using approved registration fields.
- ✓ FR-AUTH-002: The system shall validate required fields.
- ✓ FR-AUTH-003: The system shall prevent duplicate accounts according to the approved account-identification policy.
- ✓ FR-AUTH-004: The system shall allow registered users to log in.
- ✓ FR-AUTH-005: The system shall allow authenticated users to log out.
- ✓ FR-AUTH-006: The system shall store passwords using a modern password-hashing algorithm.
- ✓ FR-AUTH-007: The system shall support secure password recovery.
- ✓ FR-AUTH-008: The system shall enforce role-based access control.
- ✓ FR-AUTH-009: The system shall restrict suspended accounts.
- ✓ FR-AUTH-010: The system shall provide safe authentication error messages.
- ✓ FR-AUTH-011: The system shall manage authentication expiration and invalidation securely.
- ✓ FR-AUTH-012: Users shall be able to update permitted profile information.
- ✓ FR-AUTH-013: Support Agents shall not be allowed to retrieve user passwords.
- ✓ FR-AUTH-014: Only authorized administrators shall change privileged roles.
- ✓ FR-AUTH-015: Security-relevant authentication events should be recorded.

5.2 Farmer Profile Management

Feature ID: FEAT-FARMER

- ➢ FR-FARMER-001: The system shall maintain a farmer profile.

- FR-FARMER-002: The profile shall contain approved personal and agricultural information.
- FR-FARMER-003: The system shall record the farmer's preferred access method where appropriate.
- FR-FARMER-004: The system shall allow farmers to update permitted profile fields.
- FR-FARMER-005: The system shall maintain farmer location information where provided.
- FR-FARMER-006: The system shall maintain the farmer's product listings.
- FR-FARMER-007: The system shall display the farmer's order history.
- FR-FARMER-008: The system shall display relevant consultation history.
- FR-FARMER-009: The system shall record profile verification status where required.
- FR-FARMER-010: The system shall prevent unauthorized access to private farmer information.

5.3 Assisted Farmer Onboarding

Feature ID: FEAT-ASSISTED

This feature supports traditional farmers who cannot independently use a smartphone or web application.

- FR-ASSISTED-001: The system shall allow authorized Support Agents to initiate farmer registration.
- FR-ASSISTED-002: The system shall allow Support Agents to search for an existing farmer account before creating a new account.
- FR-ASSISTED-003: The system shall record the Support Agent responsible for assisted registration.
- FR-ASSISTED-004: The system shall record the farmer's consent before creating a listing or performing another consent-dependent action.
- FR-ASSISTED-005: The system shall record the type and time of consent.
- FR-ASSISTED-006: The system shall allow an agent to create a farmer listing on behalf of the farmer.

- FR-ASSISTED-007: The system shall preserve the farmer as the listing owner.
- FR-ASSISTED-008: The system shall record the Support Agent as the person who performed the assisted action.
- FR-ASSISTED-009: The system shall allow the farmer or an authorized representative to review listing details before publication.
- FR-ASSISTED-010: The system shall preserve a record of assisted listing changes.
- FR-ASSISTED-011: The system shall allow a farmer to request correction of assisted information.
- FR-ASSISTED-012: The system shall allow the farmer to withdraw consent for future assisted actions.
- FR-ASSISTED-013: The system shall provide administrators with assisted-service reports.
- FR-ASSISTED-014: The system shall prevent Support Agents from using a farmer's password or verification credentials.
- FR-ASSISTED-015: The system should support an assisted-access record that can be reviewed by the farmer through a support point or other approved communication channel.

5.4 Agricultural Product Marketplace

Feature ID: FEAT-MARKET

- FR-MARKET-001: The system shall allow authorized farmers to create agricultural product listings.
- FR-MARKET-002: A listing shall include a product name, category, description, quantity, unit, price, location, availability, and farmer owner.
- FR-MARKET-003: The system should allow product-image uploads.
- FR-MARKET-004: Farmers shall be able to edit their own listings.
- FR-MARKET-005: Farmers shall be able to deactivate unavailable products.
- FR-MARKET-006: The system shall support draft, pending review, published, rejected, inactive, and sold-out statuses.
- FR-MARKET-007: Users shall be able to search agricultural products.

- FR-MARKET-008: The system shall support product filtering.
- FR-MARKET-009: The system should support sorting.
- FR-MARKET-010: The system shall display permitted product details.
- FR-MARKET-011: The backend shall verify listing ownership before changes.
- FR-MARKET-012: The system shall reject invalid quantities.
- FR-MARKET-013: The system shall reject negative or invalid prices.
- FR-MARKET-014: The system shall manage available quantities according to order rules.
- FR-MARKET-015: Users shall be able to report suspicious listings.
- FR-MARKET-016: Administrators shall be able to moderate listings.
- FR-MARKET-017: The system shall display current recorded availability.
- FR-MARKET-018: The system should preserve relevant listing history.
- FR-MARKET-019: The system shall identify whether a listing was created directly by a farmer or through assisted support.
- FR-MARKET-020: The system shall display the farmer as the seller regardless of who entered the listing.

5.5 Agricultural Input Marketplace

Feature ID: FEAT-INPUT

- FR-INPUT-001: The system shall allow authorized Input Suppliers to create input listings.
- FR-INPUT-002: Supported categories may include seeds, fertilizers, tools, and agricultural equipment.
- FR-INPUT-003: Input listings shall contain supplier, name, description, category, price, unit, and availability.
- FR-INPUT-004: Buyers shall be able to search and filter input listings.
- FR-INPUT-005: Suppliers shall manage their own input listings.
- FR-INPUT-006: Administrators shall moderate reported input listings.
- FR-INPUT-007: The system shall not automatically certify that a product is genuine, safe, or legally approved without a supporting verification process.
- FR-INPUT-008: The project owner shall define any required supplier documentation.

5.6 Search and Discovery

Feature ID: FEAT-SEARCH

- FR-SEARCH-001: Users shall be able to search agricultural products by name.
- FR-SEARCH-002: Users shall be able to filter by category.
- FR-SEARCH-003: Users shall be able to filter by location.
- FR-SEARCH-004: Users shall be able to filter by price range.
- FR-SEARCH-005: Users shall be able to filter by availability.
- FR-SEARCH-006: Users shall be able to view product details.
- FR-SEARCH-007: Search results shall show the latest recorded listing information.
- FR-SEARCH-008: The system should support pagination.
- FR-SEARCH-009: The system should support relevant sorting options.

5.7 Shopping Cart and Order Management

Feature ID: FEAT-ORDER

- FR-ORDER-001: The system shall allow authenticated buyers to create orders for available products.
- FR-ORDER-002: Orders shall record the buyer, seller, products, quantities, prices, totals, and timestamps.
- FR-ORDER-003: The system shall preserve the agreed price at the time of order creation.
- FR-ORDER-004: The system shall support an order-confirmation workflow.
- FR-ORDER-005: Sellers shall be able to accept or reject eligible pending orders.
- FR-ORDER-006: The system shall support order statuses such as pending, confirmed, preparing, completed, cancelled, and disputed.
- FR-ORDER-007: The backend shall validate order-status transitions.
- FR-ORDER-008: Users shall view only orders in which they are authorized participants.
- FR-ORDER-009: The system shall allow cancellation according to the approved policy.
- FR-ORDER-010: The system shall prevent inventory overselling.
- FR-ORDER-011: The backend shall calculate order totals.
- FR-ORDER-012: The system shall notify relevant users of important order changes.

- FR-ORDER-013: The system shall preserve completed order records.
- FR-ORDER-014: The system shall not create or manage transportation bookings.
- FR-ORDER-015: The system shall not claim that an order has been physically delivered unless an approved process records that information.

5.8 Agricultural Consultation Management

Feature ID: FEAT-CONSULT

- FR-CONSULT-001: Farmers shall be able to browse available agricultural experts.
- FR-CONSULT-002: Expert profiles shall show permitted professional information.
- FR-CONSULT-003: Farmers shall be able to submit consultation requests.
- FR-CONSULT-004: Requests shall record the farmer, expert, subject, description, and requested time.
- FR-CONSULT-005: Experts shall be able to accept or reject assigned requests.
- FR-CONSULT-006: Experts shall be able to reschedule requests.
- FR-CONSULT-007: The system shall maintain consultation status.
- FR-CONSULT-008: Experts shall be able to record relevant consultation outcomes.
- FR-CONSULT-009: Farmers shall be able to view their consultation history.
- FR-CONSULT-010: The system shall notify relevant users about consultation updates.
- FR-CONSULT-011: The system shall support consultation feedback.
- FR-CONSULT-012: Consultation advice shall be provided by human experts.

5.9 Market Information Management

Feature ID: FEAT-PRICE

- FR-PRICE-001: Authorized administrators shall be able to publish market-price information.
- FR-PRICE-002: A price record shall include product, market location, unit, price, source, and observation date where available.
- FR-PRICE-003: Users shall be able to browse market information.
- FR-PRICE-004: Users shall be able to filter information by product and location.

- FR-PRICE-005: The system shall display the date associated with a market-price record.
- FR-PRICE-006: The system shall distinguish published market information from a seller's actual offer price.
- FR-PRICE-007: The system shall not present unverified data as official market information.
- FR-PRICE-008: Administrators shall be able to update or deactivate outdated information.

5.10 Educational Articles and Resources

Feature ID: FEAT-ARTICLE

- FR-ARTICLE-001: Administrators shall be able to create educational articles.
- FR-ARTICLE-002: Articles shall include a title, category, content, author or source, and publication status.
- FR-ARTICLE-003: The system shall support article images where appropriate.
- FR-ARTICLE-004: Users shall be able to browse published articles.
- FR-ARTICLE-005: Users shall be able to search articles.
- FR-ARTICLE-006: Users shall be able to filter articles by category.
- FR-ARTICLE-007: The system should support English and Amharic content.
- FR-ARTICLE-008: Administrators shall be able to update and deactivate articles.

5.11 Messaging

Feature ID: FEAT-MESSAGE

- FR-MESSAGE-001: The system shall support authorized communication between relevant users.
- FR-MESSAGE-002: Messages shall be associated with the sender and recipient.
- FR-MESSAGE-003: The system shall record message timestamps.
- FR-MESSAGE-004: Users shall only access conversations in which they are authorized participants.
- FR-MESSAGE-005: The system should support message-read status.
- FR-MESSAGE-006: Users shall be able to report inappropriate messages.

- FR-MESSAGE-007: Support Agents shall only access conversations when authorized by the applicable support workflow.

5.12 Notifications

Feature ID: FEAT-NOTIFICATION

- FR-NOTIFY-001: The system shall notify users about important account events.
- FR-NOTIFY-002: The system shall notify farmers about incoming orders.
- FR-NOTIFY-003: The system shall notify buyers about order updates.
- FR-NOTIFY-004: The system shall notify experts about consultation requests.
- FR-NOTIFY-005: The system shall notify users about consultation updates.
- FR-NOTIFY-006: The system shall notify users about support-ticket updates.
- FR-NOTIFY-007: The system should support in-app notifications.
- FR-NOTIFY-008: SMS or email notifications may be integrated after approval.
- FR-NOTIFY-009: The system shall record notification status.
- FR-NOTIFY-010: Notifications shall not expose sensitive private information.

For traditional farmers, Support Agents may help communicate important notifications through the approved assisted-support process.

5.13 Support Ticket Management

Feature ID: FEAT-SUPPORT

- FR-SUPPORT-001: Users shall be able to submit support tickets.
- FR-SUPPORT-002: Tickets shall record the user, subject, description, category, priority, and status.
- FR-SUPPORT-003: Support Agents shall be able to view assigned tickets.
- FR-SUPPORT-004: Support Agents shall be able to respond to assigned tickets.
- FR-SUPPORT-005: Support Agents shall be able to escalate unresolved issues.
- FR-SUPPORT-006: Administrators shall be able to assign and monitor tickets.
- FR-SUPPORT-007: The system shall preserve ticket history.

- FR-SUPPORT-008: Users shall be able to view their own ticket status.
- FR-SUPPORT-009: The system shall record important support actions.

5.14 Reviews and Ratings

Feature ID: FEAT-REVIEW

- FR-REVIEW-001: Eligible users shall be able to submit reviews.
- FR-REVIEW-002: Reviews shall be associated with the relevant user and transaction where applicable.
- FR-REVIEW-003: The system shall validate review eligibility.
- FR-REVIEW-004: Administrators shall be able to moderate reported reviews.
- FR-REVIEW-005: The system shall prevent unauthorized users from modifying other users' reviews.

5.15 Complaints and Dispute Management

Feature ID: FEAT-DISPUTE

- FR-DISPUTE-001: Users shall be able to submit transaction-related complaints.
- FR-DISPUTE-002: Disputes shall record the complainant, relevant transaction, reason, description, and status.
- FR-DISPUTE-003: Support Agents shall be able to receive and escalate complaints.
- FR-DISPUTE-004: Administrators shall review assigned disputes.
- FR-DISPUTE-005: The system shall preserve dispute history.
- FR-DISPUTE-006: The system shall record the responsible administrator and resolution.
- FR-DISPUTE-007: Users shall be able to view permitted dispute updates.

5.16 Administrator Management

Feature ID: FEAT-ADMIN

- FR-ADMIN-001: Administrators shall be able to manage users.
- FR-ADMIN-002: Administrators shall manage role assignments.

- FR-ADMIN-003: Administrators shall review verification requests.
- FR-ADMIN-004: Administrators shall manage Support Agent accounts.
- FR-ADMIN-005: Administrators shall monitor assisted farmer activity.
- FR-ADMIN-006: Administrators shall moderate agricultural listings.
- FR-ADMIN-007: Administrators shall manage agricultural categories.
- FR-ADMIN-008: Administrators shall manage articles.
- FR-ADMIN-009: Administrators shall manage market-price information.
- FR-ADMIN-010: Administrators shall monitor orders.
- FR-ADMIN-011: Administrators shall monitor consultation activities.
- FR-ADMIN-012: Administrators shall manage complaints and disputes.
- FR-ADMIN-013: Administrators shall generate reports.
- FR-ADMIN-014: Administrators shall review audit logs.
- FR-ADMIN-015: The system shall record privileged administrative actions.

6. Detailed Use Cases and Workflows

6.1 Use Case: Direct Farmer Registration

Primary Actor: Farmer

Precondition: The farmer has access to the web or mobile application.

Main Flow:

1. Farmer opens AgroLink.
2. Farmer selects registration.
3. Farmer enters required information.
4. Farmer selects Farmer as the account type.
5. System validates the information.
6. System creates the account.
7. Farmer completes any required verification.
8. Farmer logs in.
9. System displays the Farmer Dashboard.

Postcondition: A farmer account is available.

6.2 Use Case: Assisted Farmer Registration

Primary Actor: Support Agent

Beneficiary: Farmer

Precondition: Support Agent is authenticated and authorized.

Main Flow:

1. Support Agent opens the assisted farmer-registration interface.
2. Agent searches for an existing farmer account.
3. System checks for possible duplicates.
4. Agent collects the required farmer information.
5. Agent explains the account and support process.
6. Farmer provides the required consent.
7. Agent records the consent.
8. System creates the farmer account.
9. System records the Support Agent responsible.
10. System displays the registration confirmation.

Alternative Flow:

- If the farmer already exists, the system displays the existing account according to access permissions.
- If consent is not provided, the system shall not perform the consent-dependent registration or listing action.
- If required information is missing, the system shall request correction.

Postcondition: A farmer account is created with an auditable assisted-registration record.

6.3 Use Case: Direct Farmer Creates Product

Primary Actor: Farmer

Main Flow:

1. Farmer logs in.
2. Farmer opens My Products.
3. Farmer selects Add Product.
4. Farmer enters product details.
5. Farmer uploads product images where available.
6. Farmer reviews the information.
7. Farmer submits the listing.
8. System validates the listing.
9. System stores the listing under the farmer's account.
10. Listing enters the applicable publication workflow.

Postcondition: A farmer-owned listing is stored.

6.4 Use Case: Support Agent Creates Product on Behalf of Farmer

Primary Actor: Support Agent

Beneficiary: Farmer

Main Flow:

1. Support Agent logs in.
2. Agent searches for the farmer.
3. Agent opens the farmer's account using the authorized assisted workflow.
4. Agent confirms the farmer's consent.
5. Agent opens Create Product on Behalf of Farmer.
6. Agent enters the product information provided by the farmer.
7. Agent reviews the product details with the farmer.
8. Farmer confirms the details.

9. Agent submits the listing.
10. System records the farmer as owner.
11. System records the Support Agent as the creator of the assisted action.
12. Listing enters the applicable publication workflow.

Postcondition: A product listing exists under the farmer's ownership with an associated assisted-action record.

6.5 Use Case: Buyer Places an Order

Primary Actor: Buyer

Main Flow:

1. Buyer logs in.
2. Buyer searches the marketplace.
3. Buyer opens product details.
4. Buyer selects quantity.
5. System validates availability.
6. Buyer confirms the order.
7. System calculates the order total.
8. System creates the order.
9. System notifies the seller.
10. Seller reviews and responds.
11. System updates the order status.
12. Buyer views the updated status.

Postcondition: An order is created and recorded.

6.6 Use Case: Agricultural Consultation

Primary Actor: Farmer

Supporting Actor: Agricultural Expert

Main Flow:

1. Farmer opens the Expert Directory.
2. Farmer views expert profiles.
3. Farmer selects an expert.
4. Farmer submits a consultation request.
5. System records the request.
6. Expert receives a notification.
7. Expert accepts, rejects, or proposes a new time.
8. Farmer receives the response.
9. Consultation takes place through the agreed method.
10. Expert records the consultation status.
11. Farmer may submit feedback.

Postcondition: The consultation record is maintained.

6.7 Use Case: Support Agent Assists with an Order

Primary Actor: Support Agent

Beneficiary: Farmer

Main Flow:

1. Farmer contacts the Support Agent.
2. Agent verifies the farmer using the approved process.
3. Agent opens the farmer's relevant order information.
4. Agent explains the order status.
5. Agent records assistance if required.
6. Agent escalates an issue when necessary.

The Support Agent does not automatically accept, reject, or cancel an order for the farmer.

6.8 Use Case: Administrator Reviews Assisted Activity

Primary Actor: Administrator

Main Flow:

1. Administrator logs in.
2. Administrator opens Assisted Farmer Management.
3. Administrator selects an assisted action.
4. System displays farmer, agent, consent, action, and timestamp.
5. Administrator reviews the record.
6. Administrator may investigate suspicious activity.
7. System records the administrative review.

7. Business Rules and Authorization Policies

7.1 General Business Rules

- BR-001: Every registered account shall have an authorized account type.
- BR-002: A user shall access only permitted features.
- BR-003: Every farmer listing shall belong to a farmer account.
- BR-004: Buyers shall not modify farmer listings.
- BR-005: Input Suppliers shall manage only their own input listings.
- BR-006: Agricultural Experts shall access only authorized consultation records.
- BR-007: Support Agents shall only perform approved support activities.
- BR-008: Administrators shall have privileged access according to approved policies.
- BR-009: Important actions shall be recorded in audit logs.
- BR-010: Order totals shall be calculated by the backend.
- BR-011: Product quantities and prices shall be validated.
- BR-012: Historical order records shall be preserved.
- BR-013: Transportation management shall not be included in this release.
- BR-014: Agricultural advice shall be provided by human experts.
- BR-015: Market information shall include a source and date where available.

7.2 Assisted Farmer Business Rules

- BR-016: A Support Agent shall not create a farmer listing without the required farmer consent.
- BR-017: The farmer shall remain the owner of the listing.
- BR-018: The Support Agent shall be recorded as the person performing the assisted action.
- BR-019: The system shall record the consent type and timestamp.
- BR-020: A Support Agent shall not access the farmer's password or one-time verification code.
- BR-021: A Support Agent shall not independently decide the farmer's selling price.
- BR-022: The farmer shall be given an opportunity to confirm important listing information.
- BR-023: An agent's permission shall be limited to assigned and authorized support activities.
- BR-024: A farmer shall be able to request corrections to assisted information.
- BR-025: The system shall maintain a traceable history of assisted changes.

7.3 Service Provider Rules

- BR-026: A Service Provider account shall have an approved provider type.
- BR-027: Input Suppliers shall access supplier-specific functions.
- BR-028: Agricultural Experts shall access expert-specific functions.
- BR-029: A provider shall not automatically receive all permissions belonging to other provider types.

7.4 Order Rules

- BR-030: Buyers shall order only available products.
- BR-031: Order quantities shall not exceed available quantities.
- BR-032: Sellers shall manage orders related to their own products.
- BR-033: Order status changes shall follow approved transitions.
- BR-034: The system shall preserve the price agreed at order creation.

- BR-035: Cancelled and completed order records shall be retained according to the approved retention policy.

8. Non-Functional Requirements

8.1 Performance

- NFR-PERF-001: Common pages should load within an acceptable time under normal network conditions.
- NFR-PERF-002: APIs should respond efficiently under the agreed test workload.
- NFR-PERF-003: Product search should use appropriate database indexing and pagination.
- NFR-PERF-004: Large image files shall be restricted and optimized.
- NFR-PERF-005: The backend should support concurrent users without corrupting order or inventory records.

Exact performance targets shall be agreed and tested before final acceptance.

8.2 Usability

- NFR-USE-001: The interface shall use clear navigation.
- NFR-USE-002: Common actions shall have understandable labels.
- NFR-USE-003: Farmer workflows shall avoid unnecessary complexity.
- NFR-USE-004: The assisted-support interface shall be simple for Support Agents.
- NFR-USE-005: The mobile interface shall be designed for smaller screens.
- NFR-USE-006: Important confirmation and error messages shall be understandable.
- NFR-USE-007: The interface should use consistent colors, typography, spacing, and components.

8.3 Accessibility and Inclusion

- NFR-ACCESS-001: The interface should follow practical WCAG accessibility guidance.
- NFR-ACCESS-002: Text shall be readable on mobile devices.
- NFR-ACCESS-003: Important actions shall not depend only on color.

- NFR-ACCESS-004: Forms shall have clear labels.
- NFR-ACCESS-005: The design shall support users with different levels of digital literacy.
- NFR-ACCESS-006: English and Amharic localization shall be supported.
- NFR-ACCESS-007: Assisted farmer workflows shall not require the farmer to personally operate a smartphone.

8.4 Reliability

- NFR-REL-001: The system shall validate important operations before saving.
- NFR-REL-002: The database shall use appropriate constraints and transactions.
- NFR-REL-003: The system shall prevent duplicate order submissions where practical.
- NFR-REL-004: Failed operations shall return controlled errors.
- NFR-REL-005: Backup and restoration procedures shall be documented and tested.

8.5 Security

- NFR-SEC-001: All production traffic shall use HTTPS.
- NFR-SEC-002: Passwords shall be securely hashed.
- NFR-SEC-003: Authentication and authorization shall be enforced on the backend.
- NFR-SEC-004: Users shall not access records belonging to unauthorized users.
- NFR-SEC-005: API inputs shall be validated.
- NFR-SEC-006: Database queries shall use parameterized queries or a secure ORM approach.
- NFR-SEC-007: Secrets shall be stored in environment configuration or an approved secret manager.
- NFR-SEC-008: Rate limiting shall protect sensitive endpoints.
- NFR-SEC-009: File uploads shall be validated.
- NFR-SEC-010: Important actions shall be recorded in audit logs.
- NFR-SEC-011: The system shall protect against common web and API security vulnerabilities.
- NFR-SEC-012: Support Agents shall not access user passwords or authentication secrets.
- NFR-SEC-013: Assisted farmer records shall be protected from unauthorized access.

8.6 Maintainability

- NFR-MAINT-001: Frontend and backend code shall be modular.
- NFR-MAINT-002: Business logic shall be separated from HTTP controllers.
- NFR-MAINT-003: Database changes shall use versioned migrations.
- NFR-MAINT-004: API documentation shall be maintained.
- NFR-MAINT-005: Reusable UI components shall be used where practical.
- NFR-MAINT-006: Code shall follow consistent naming and formatting conventions.

8.7 Scalability

- NFR-SCALE-001: The backend should support additional users and modules.
- NFR-SCALE-002: The architecture should allow additional mobile and web features without redesigning the entire system.
- NFR-SCALE-003: Database queries shall be optimized as data volume increases.
- NFR-SCALE-004: Image storage should be separated from database records where appropriate.

8.8 Compatibility

- NFR-COMP-001: The web application shall support modern browsers.
- NFR-COMP-002: The mobile application shall support the approved Android versions.
- NFR-COMP-003: The mobile application shall use the same backend API as the web application.
- NFR-COMP-004: The system shall provide consistent business rules across both clients.

9. System Architecture and Technology Requirements

9.1 Recommended Technology Stack

Layer	**Technology**
Web Frontend	React.js

Web Build Tool	Vite
Web Language	JavaScript / JSX
Mobile Application	React Native
Mobile Navigation	React Navigation
Backend	Node.js
Backend Framework	Express.js
API Style	REST
Database	PostgreSQL
Authentication	Secure session or token-based authentication
Password Hashing	bcrypt or an approved equivalent
Image Storage	Approved object-storage service
API Testing	Postman
Frontend Testing	Vitest and React Testing Library
Backend Testing	Jest or an approved runner, Supertest
Version Control	Git and GitHub

9.2 Web and Mobile Architecture Decision

The recommended architecture is to maintain two frontend clients:

- React + Vite for the web application.
- React Native for the Android/iOS mobile application.

Both clients shall use one shared backend API and PostgreSQL database.

This is a practical choice because React Native and browser-based React/Vite have different rendering and build environments.

Shared business rules, validation definitions, API service patterns, and design tokens may be reused where appropriate. The frontend interfaces shall still be designed for their respective platforms.

If the project later chooses Expo with React Native Web to share more interface code, that should be treated as a deliberate architecture decision rather than mixing Vite and React Native build configurations without a clear design.

9.3 Backend Architecture

The backend should use a modular structure:

1. Routes.
2. Controllers.
3. Services.
4. Repositories or database-access modules.
5. Models or schemas.
6. Middleware.
7. Validators.
8. Policies.
9. Integrations.
10. Tests.

9.4 Authentication Architecture

The backend shall authenticate users and determine their roles and permissions.

The web and mobile clients shall not independently decide whether a user is authorized to perform a protected operation.

The backend shall validate the authenticated user, requested action, resource ownership, and role permissions.

9.5 Assisted Access Architecture

The assisted-access module shall provide a controlled workflow for Support Agents.

The backend shall record:

- Farmer account ID.
- Support Agent account ID.
- Action type.
- Consent status.
- Consent method.
- Consent timestamp.
- Listing or record affected.
- Previous and updated values where relevant.
- Action result.

10. Database Requirements and Data Model

10.1 Database Management System

The system shall use PostgreSQL as its relational database.

The database shall maintain relationships between users, profiles, listings, orders, consultations, support activities, and administrative records.

10.2 Main Database Entities

The proposed database shall include the following entities:

Entity	**Purpose**
users	Stores authentication and account information
roles	Stores the five main role definitions
user_profiles	Stores user profile information
farmer_profiles	Stores farmer-specific information
service_provider_profiles	Stores provider information
provider_types	Identifies input supplier or agricultural expert
support_agent_profiles	Stores Support Agent information
assisted_farmer_actions	Records assisted activities

farmer_consents	Stores consent records
product_categories	Stores agricultural categories
products	Stores farmer agricultural products
product_images	Stores product image references
input_products	Stores agricultural input listings
input_categories	Stores input categories
orders	Stores buyer orders
order_items	Stores ordered products and quantities
consultations	Stores agricultural consultation requests
expert_availability	Stores expert availability
market_prices	Stores market-price information
articles	Stores educational articles
article_categories	Stores article categories
conversations	Stores conversation records
messages	Stores user messages
notifications	Stores user notifications
support_tickets	Stores support requests
support_ticket_messages	Stores support-ticket communication
reviews	Stores eligible user reviews
disputes	Stores complaints and disputes
audit_logs	Stores important system actions

10.3 Important Database Relationships

- One user may have one profile.
- One farmer may own many agricultural products.
- One Support Agent may perform many assisted actions.
- One farmer may have many assisted actions.
- One assisted action may be associated with a farmer consent record.
- One input supplier may publish many input products.
- One agricultural expert may receive many consultation requests.

- One buyer may create many orders.
- One order may contain multiple order items.
- One product may appear in many historical order items.
- One user may receive many notifications.
- One support ticket may contain many messages.
- One user may submit eligible reviews.
- One administrator may create many audit-log entries.

10.4 Assisted Listing Data Requirements

A farmer product listing shall contain:

- Listing ID.
- Farmer owner ID.
- Product name.
- Product category.
- Description.
- Quantity.
- Unit.
- Price.
- Location.
- Availability.
- Listing status.
- Created timestamp.
- Updated timestamp.
- Created-by user ID.
- Creation method.

The creation method should distinguish:

- DIRECT_FARMER
- ASSISTED_SUPPORT

An assisted-action record shall contain:

- Assisted action ID.
- Farmer ID.
- Support Agent ID.
- Action type.
- Related listing ID where applicable.
- Consent record ID.
- Timestamp.
- Action status.
- Relevant audit information.

The system shall never replace the farmer owner ID with the Support Agent ID merely because the agent entered the data.

10.5 Database Integrity

The database shall:

- Use primary keys.
- Use foreign keys.
- Enforce required fields.
- Enforce appropriate uniqueness constraints.
- Validate important values.
- Use transactions for sensitive multi-step operations.
- Preserve historical order records.
- Use migrations for schema changes.
- Restrict database access through approved application credentials.

11. Web and Mobile Frontend Requirements

11.1 Public Web Pages

The web application should include:

- Home Page.
- About AgroLink.
- Marketplace.
- Product Details.
- Agricultural Inputs.
- Market Prices.
- Agricultural Articles.
- Expert Directory.
- Login.
- Registration.
- Contact and Support.

11.2 Farmer Web Dashboard

The Farmer Dashboard should include:

- Overview.
- My Products.
- Add Product.
- Product Management.
- Orders.
- Consultation Requests.
- Market Prices.
- Articles and Resources.
- Messages.
- Notifications.
- Transaction History.
- Support.
- Profile.
- Settings.

11.3 Buyer Dashboard

The Buyer Dashboard should include:

- Overview.
- Marketplace.
- Search Products.
- Cart.
- My Orders.
- Order Details.
- Messages.
- Notifications.
- Reviews.
- Disputes.
- Support.
- Profile.
- Settings.

11.4 Service Provider Dashboard

The Service Provider interface shall depend on provider type.

Input Supplier pages:

- Overview.
- My Input Products.
- Add Input Product.
- Inventory.
- Orders.
- Customer Messages.
- Reviews.
- Notifications.
- Profile.

- Settings.

Agricultural Expert pages:

- Overview.
- Professional Profile.
- Expertise.
- Availability.
- Consultation Requests.
- Consultation History.
- Messages.
- Feedback.
- Notifications.
- Settings.

11.5 Support Agent Dashboard

The Support Agent Dashboard should include:

- Overview.
- Farmer Search.
- Assisted Farmer Registration.
- Farmer Profiles.
- Create Product on Behalf of Farmer.
- Assisted Product Listings.
- Consent Records.
- Order Assistance.
- Consultation Assistance.
- Support Tickets.
- Assigned Tasks.
- Escalated Complaints.
- Assistance History.
- Profile.

- Settings.

The interface shall clearly show when the agent is acting on behalf of a farmer.

11.6 Administrator Dashboard

The Administrator Dashboard should include:

- Overview.
- User Management.
- Role and Permission Management.
- Farmer Management.
- Support Agent Management.
- Assisted Farmer Monitoring.
- Consent and Assistance Records.
- Buyer Management.
- Service Provider Management.
- KYC and Verification.
- Product Management.
- Input Supplier Management.
- Expert Management.
- Market Price Management.
- Educational Content.
- Order Management.
- Consultation Monitoring.
- Support Tickets.
- Dispute Management.
- Reports.
- Audit Logs.
- System Settings.

11.7 Mobile Application Requirements

The mobile application shall provide role-appropriate interfaces.

The Farmer mobile experience should emphasize:

- Simple registration and login.
- Clear product creation.
- Product image upload.
- Product and order status.
- Consultation requests.
- Market information.
- Notifications.
- Messages.
- Profile management.

The Buyer mobile experience should emphasize:

- Product discovery.
- Search and filtering.
- Product details.
- Cart and order creation.
- Order tracking within the platform.
- Messages and notifications.

Service Provider mobile screens shall be determined by provider type.

The mobile application shall not include transporter or delivery-management screens in this release.

11.8 UI/UX Design Principles

The interface should:

- Use clear and consistent navigation.
- Use readable typography.

- Provide clear form validation.
- Use consistent buttons and cards.
- Support responsive layouts.
- Minimize unnecessary steps.
- Provide confirmation before important changes.
- Support English and Amharic text.
- Use a simple visual design suitable for a broad range of digital-literacy levels.

12. Backend and REST API Requirements

12.1 API Architecture

The backend shall expose RESTful APIs.

The web and mobile applications shall consume the same API endpoints wherever possible.

All protected endpoints shall require valid authentication and authorization.

12.2 Suggested API Modules

Module	**Example API Prefix**
Authentication	/api/v1/auth
Users	/api/v1/users
Farmer Profiles	/api/v1/farmers
Assisted Farmer	/api/v1/assisted-farmers
Farmer Products	/api/v1/products
Product Categories	/api/v1/categories
Input Products	/api/v1/inputs
Orders	/api/v1/orders
Service Providers	/api/v1/service-providers
Experts	/api/v1/experts
Consultations	/api/v1/consultations

Market Prices	/api/v1/market-prices
Articles	/api/v1/articles
Messages	/api/v1/messages
Notifications	/api/v1/notifications
Support Tickets	/api/v1/support-tickets
Reviews	/api/v1/reviews
Disputes	/api/v1/disputes
Reports	/api/v1/reports
Audit Logs	/api/v1/audit-logs

12.3 Example Assisted Farmer Endpoints

- POST /api/v1/assisted-farmers/register
- GET /api/v1/assisted-farmers/search
- POST /api/v1/assisted-farmers//consents
- POST /api/v1/assisted-farmers//products
- GET /api/v1/assisted-farmers//history
- PATCH /api/v1/assisted-farmers//products/
- GET /api/v1/assisted-farmers/reports

The backend shall verify that the Support Agent is authorized for each operation.

12.4 API Response Requirements

API responses should use a consistent structure.

A successful response should include:

- Success status.
- Message.
- Data.
- Relevant pagination metadata where required.

An error response should include:

- Error status.
- Safe message.
- Error code.
- Field validation details where applicable.

The API shall not expose stack traces, secrets, passwords, or sensitive internal information to clients.

13. External Interfaces and Integrations

13.1 Payment Integration

The first release may record payment status without integrating a live payment gateway.

If an online payment provider is selected later, the integration shall require:

- Project-owner approval.
- Provider documentation.
- Secure server-side verification.
- Transaction references.
- Error handling.
- Payment audit records.

A browser redirect alone shall not be treated as proof of successful payment.

13.2 SMS and Email

The system may use approved services for:

- Account verification.
- Password recovery.
- Important order notifications.

- Consultation updates.
- Support-ticket updates.

SMS integration shall be subject to provider availability and cost.

13.3 Image Storage

The system may use approved object storage for:

- Farmer product images.
- Input product images.
- Profile images.
- Article images.
- Approved supporting documents.

The database should store image metadata and storage references rather than unnecessarily storing large image files directly.

13.4 Maps and Location Services

The system may use approved map services for displaying user-provided market and product locations.

Map-based transport booking and delivery tracking are excluded.

14. Security, Privacy, and Data Protection

14.1 Authentication

The system shall authenticate users before granting access to protected resources.

Passwords shall be hashed and never stored as plain text.

14.2 Authorization

- The backend shall enforce role-based and resource-based authorization.
- A farmer shall only manage their own listings.
- A Support Agent shall only perform authorized assisted actions.
- An Input Supplier shall only manage their own input listings.
- An Agricultural Expert shall only manage authorized consultation records.
- An Administrator shall access privileged management functions.

14.3 Assisted Farmer Privacy

The assisted farmer workflow shall protect farmers from unauthorized actions.

The system shall:

- Record farmer consent.
- Record the responsible Support Agent.
- Record the action performed.
- Restrict agent permissions.
- Preserve listing ownership.
- Protect farmer contact information.
- Maintain auditable records.
- Support correction of inaccurate assisted information.

14.4 Application Security

The implementation shall consider protection against:

- SQL injection.
- Cross-site scripting.
- Cross-site request forgery where applicable.
- Broken access control.
- Brute-force attacks.
- Insecure file uploads.
- Sensitive data exposure.

- Insecure session management.
- API abuse.

14.5 Data Privacy

The system shall collect only information required for its approved purposes.

Private information shall be visible only to authorized users.

Data retention and deletion policies shall be defined by the project owner with regard to applicable Ethiopian requirements.

15. Error Handling and Validation

15.1 Validation

The system shall validate:

- Required fields.
- Email or phone formats where used.
- Password rules.
- Product quantities.
- Product prices.
- Order quantities.
- Image file types.
- Image file sizes.
- User permissions.
- Consent requirements.
- Consultation request fields.

15.2 Error Handling

The system shall provide understandable error messages.

Examples include:

- Invalid login credentials.
- Required information is missing.
- Product quantity is unavailable.
- Product listing was not found.
- You do not have permission to perform this action.
- Farmer consent is required.
- Consultation request could not be submitted.
- The service is temporarily unavailable.

15.3 Audit and Logging

The system should record:

- Authentication events.
- Account changes.
- Role changes.
- Listing creation and updates.
- Assisted farmer actions.
- Consent records.
- Order-status changes.
- Consultation-status changes.
- Administrative decisions.
- Dispute resolutions.

Passwords, authentication tokens, and other secrets shall not be recorded in logs.

16. Testing and Quality Assurance

16.1 Unit Testing

Unit tests shall verify individual functions and business rules.

Examples:

- Price calculations.
- Quantity validation.
- Role permission checks.
- Consent validation.
- Order status transitions.

16.2 Integration Testing

Integration tests shall verify:

- Frontend-to-backend communication.
- Backend-to-database communication.
- Authentication and authorization.
- Product creation.
- Assisted product creation.
- Order creation.
- Consultation requests.
- Notification creation.

16.3 Web Application Testing

The web application shall be tested for:

- Navigation.
- Form validation.
- Responsive layouts.
- Role-based screens.
- Browser compatibility.
- Accessibility basics.

16.4 Mobile Application Testing

The mobile application shall be tested for:

- Android device compatibility.
- Navigation.
- Authentication.
- API communication.
- Product-image uploads.
- Order creation.
- Notifications.
- Network-failure handling.
- Screen-size compatibility.

16.5 Security Testing

Security tests shall verify:

- Unauthorized access prevention.
- Listing ownership enforcement.
- Role separation.
- Support Agent permission boundaries.
- Password protection.
- Input validation.
- Secure file uploads.
- Rate limiting.

16.6 Assisted Farmer Acceptance Testing

The following cases shall be tested:

1. A Support Agent registers a new farmer.
2. The system prevents duplicate farmer registration.
3. A Support Agent records farmer consent.
4. A Support Agent creates a product on behalf of a farmer.

5. The farmer remains the listing owner.
6. The Support Agent is recorded as the acting user.
7. A Support Agent cannot create a listing without the required consent.
8. An unauthorized Support Agent cannot edit another farmer's records.
9. The Administrator can review assisted activity.
10. The farmer can request correction of inaccurate listing information.

17. Deployment, Operations, Backup, and Recovery

17.1 Deployment

The system shall support separate environments where practical:

- ✓ Development.
- ✓ Testing.
- ✓ Production.

The web application, mobile application, backend, and database shall use documented deployment configurations.

17.2 Configuration

Environment-specific settings shall be stored outside committed source code.

The project shall provide an example environment file without real secrets.

17.3 Database Backup

The PostgreSQL database shall be backed up according to the approved operational policy.

Backup restoration shall be tested.

17.4 Monitoring

The system should monitor:

- Backend availability.
- API errors.
- Database connectivity.
- Authentication failures.
- Important application failures.
- Storage usage.

17.5 Maintenance

The development team shall document:

- Installation instructions.
- Environment configuration.
- Database migrations.
- Deployment steps.
- Backup and restoration procedures.
- Troubleshooting guidance.

18. Project Implementation Plan

The project shall use an iterative and incremental development approach.

Phase 1: Requirements and Analysis

- ✓ Review and approve the SRS.
- ✓ Confirm actors and permissions.
- ✓ Validate farmer-assisted workflows.
- ✓ Identify user-interface requirements.
- ✓ Define database entities.

Phase 2: UI/UX Design

- ✓ Design the public website.
- ✓ Design the farmer dashboard.
- ✓ Design the buyer dashboard.
- ✓ Design the Service Provider dashboard.
- ✓ Design the Support Agent dashboard.
- ✓ Design the Administrator dashboard.
- ✓ Design mobile application screens.
- ✓ Validate the design with representative users where possible.

Phase 3: Database and Backend

- ✓ Create PostgreSQL database.
- ✓ Implement migrations.
- ✓ Implement authentication.
- ✓ Implement authorization.
- ✓ Implement user profiles.
- ✓ Implement assisted farmer services.
- ✓ Implement product and input modules.
- ✓ Implement orders.
- ✓ Implement consultations.
- ✓ Implement messaging and notifications.
- ✓ Implement reports and audit logs.

Phase 4: Web Application

- ✓ Implement public pages.
- ✓ Implement authentication screens.
- ✓ Implement role-based dashboards.
- ✓ Implement marketplace.
- ✓ Implement assisted farmer interface.
- ✓ Implement administrative interface.

Phase 5: Mobile Application

- ✓ Implement React Native project.
- ✓ Implement authentication.
- ✓ Implement farmer interface.
- ✓ Implement buyer interface.
- ✓ Implement Service Provider interface where required.
- ✓ Integrate backend APIs.
- ✓ Test Android devices.

Phase 6: Testing and Integration

- ✓ Unit testing.
- ✓ API testing.
- ✓ Web testing.
- ✓ Mobile testing.
- ✓ Security testing.
- ✓ Assisted farmer workflow testing.
- ✓ User acceptance testing.

Phase 7: Deployment and Documentation

- ✓ Configure production environment.
- ✓ Deploy backend.
- ✓ Deploy web application.
- ✓ Build mobile application.
- ✓ Configure database backup.
- ✓ Prepare user documentation.
- ✓ Prepare technical documentation.
- ✓ Conduct final acceptance.

19. Project Deliverables

The project shall deliver:

1. Approved Software Requirements Specification.
2. System architecture documentation.
3. Database design and ER diagram.
4. Use-case diagram.
5. Activity diagrams for key workflows.
6. Web application source code.
7. React Native mobile application source code.
8. Node.js and Express backend source code.
9. PostgreSQL database schema and migrations.
10. API documentation.
11. UI/UX design.
12. Test plan and test cases.
13. Test results.
14. Deployment documentation.
15. User guide.
16. Final project report.

20. Requirements Traceability and Acceptance Criteria

20.1 Core Acceptance Criteria

The system shall be considered ready for project acceptance when the following critical requirements have been demonstrated:

ID	Acceptance Criterion
AC-001	Users can register and log in successfully
AC-002	Backend authorization prevents unauthorized access

AC-003	Modern farmers can create their own listings
AC-004	Support Agents can register farmers through the assisted workflow
AC-005	Support Agents can create farmer listings with recorded consent
AC-006	Assisted listings remain owned by the farmer
AC-007	The system records which Support Agent performed each assisted action
AC-008	Buyers can search and view products
AC-009	Buyers can create valid orders
AC-010	Sellers can manage their own orders
AC-011	Input Suppliers can manage input listings
AC-012	Agricultural Experts can manage consultation requests
AC-013	Market information and articles can be published and viewed
AC-014	Support tickets can be created and managed
AC-015	Administrators can manage users and moderate content
AC-016	Important actions are recorded in audit logs
AC-017	The web and mobile applications use the same backend
AC-018	The system works without transportation-management modules
AC-019	Critical security and validation tests pass
AC-020	Database backup and restoration are demonstrated

20.2 Approval

The project stakeholders should review and approve this specification before implementation is treated as final.

Role	**Name**	**Signature**	**Date**
Project Supervisor			
Project Team Leader			
Project Representative			
Testing Representative			

21. Assumptions, Constraints, and Risks

21.1 Assumptions

- ✓ Farmers and buyers will provide accurate information.
- ✓ Support Agents will be trained.
- ✓ Farmers will be informed before assisted actions.
- ✓ The project will use PostgreSQL.
- ✓ The project will provide suitable hosting.
- ✓ Internet connectivity will be available for online operations.
- ✓ Agricultural experts will provide human consultations.

21.2 Constraints

- ✓ The project has a limited development period.
- ✓ The system depends on network connectivity.
- ✓ Some farmers may not have smartphones.
- ✓ Some farmers may have limited digital literacy.
- ✓ External SMS, payment, and storage services may introduce additional costs.
- ✓ The first release will not manage transportation or delivery.

21.3 Risks and Mitigation

Risk	**Mitigation**
Traditional farmers cannot use smartphones	Provide an assisted farmer workflow
Unauthorized agent activity	Enforce permissions, consent, and audit logs
Duplicate farmer accounts	Search existing accounts before registration
Incorrect product information	Require farmer confirmation and support correction
Unavailable internet	Provide clear failure handling and avoid claiming unsaved actions succeeded
Fraudulent listings	Provide moderation and reporting

Incorrect market information	Record source, date, and location
Unauthorized access	Enforce backend authorization
Mobile compatibility problems	Test on representative Android devices
External service failure	Provide graceful failure handling and documented dependencies

22. Recommended Project Folder Structure

The implementation should separate web, mobile, backend, and shared documentation.

```
agrolink-ethiopia/
├── web/                    # React + Vite web application
├── mobile/                 # React Native application
├── server/                 # Node.js + Express API
├── docs/
│   ├── SRS.md
│   ├── architecture.md
│   ├── use-cases.md
│   ├── database-design.md
│   ├── api-documentation.md
│   ├── test-plan.md
│   ├── deployment-guide.md
│   └── user-guide.md
│
├── .gitignore
├── README.md
└── package.json
```

Transportation and delivery modules are intentionally absent from this release.

23. Glossary

Term	**Definition**
AgroLink Ethiopia	Inclusive agricultural marketplace and advisory platform

Farmer	Agricultural producer who owns and publishes agricultural products
Traditional Farmer	Farmer who requires assistance to access digital services
Modern Farmer	Farmer who can independently use the web or mobile application
Buyer	User who searches for and purchases agricultural products
Service Provider	General actor representing Input Suppliers and Agricultural Experts
Input Supplier	Provider selling agricultural inputs
Agricultural Expert	Qualified human provider of agricultural consultation
Support Agent	Authorized user who assists farmers and other platform users
Assisted Farmer Service	Workflow for helping farmers use AgroLink through authorized agents
Farmer Consent	Recorded permission for an approved assisted action
Assisted Listing	Product listing created or updated by an agent on behalf of a farmer
Listing Owner	Farmer who owns the agricultural product listing
Order	Recorded request to purchase one or more products
Consultation	Request for human agricultural guidance
Market Price	Published agricultural price information with a source and date where available
REST API	Interface used by web and mobile clients to communicate with the backend
RBAC	Role-Based Access Control
Audit Log	Record of important system actions

Final Project Statement

AgroLink Ethiopia is specified as an inclusive web and mobile agricultural marketplace and advisory platform designed for the Ethiopian agricultural environment.

The platform shall use five main actors: Farmer, Buyer, Service Provider, Support Agent, and Administrator.

It shall support both modern farmers who independently use the web or mobile application and traditional farmers who receive assistance from authorized Support Agents.

A central design principle is that the farmer remains the owner of agricultural products even when a Support Agent creates or updates a listing on the farmer's behalf. Consent, ownership, and agent accountability shall be enforced through the system's business rules and audit records.

The Service Provider actor shall represent Input Suppliers and Agricultural Experts, with separate provider types and permissions.

The first release shall include agricultural product listings, agricultural input listings, order management, consultations, market information, educational content, messaging, notifications, support services, reviews, disputes, and administration.

Transportation management, delivery tracking, transporter accounts, AI, machine learning, automated crop diagnosis, and AI-generated agricultural advice are outside the scope of this release.

The technical implementation shall use React and Vite for the web application, React Native for the mobile application, Node.js and Express.js for the backend, and PostgreSQL for the database.

The system shall use one centralized backend and database to provide consistent information and business rules across web and mobile clients.